

AskBIT / CampusIQ AI FAQ Portal

Faculty Presentation & Complete FAQ System Architecture Guide

1. Executive Summary & Core Objective

AskBIT (CampusIQ) is an AI-powered, retrieval-augmented generation (RAG) college FAQ chatbot platform designed for Bannari Amman Institute of Technology (BIT). It delivers instant, verified answers for academic regulations, fee deadlines, hostel rules, exam timetables, and campus placements.

2. Role-Based Access Control (RBAC) Roster

Role	Access Scope	Configured User Accounts	Key Responsibilities
Student	Student Support Access	1. Rahul K (AIDS) 2. Sanjiv (AIDS) 3. Sujan (IT)	Ask AI FAQ search, view 75% attendance criteria, check placement eligibility.
Parent	Linked Ward Scope	1. Kanagaraj (Rahul K) 2. Venkatachalam (Sanjiv) 3. Palanisamy (Sujan)	Track ward attendance & fees, hostel gate timings, report AI answer issues.
Faculty	Dept Intelligence Scope	1. Dr. Arun Kumar (Associate Prof, AIDS) 2. Dr. Sathishkumar (Prof, CSE)	Review assigned low-confidence queries, compose official FAQ answer drafts.
Admin	Full System Control	System Admin (admin / @min)	Manage users, assign faculty to answer FAQs, publish FAQs to vector DB.

3. Page-by-Page FAQ Portal Navigation & Features

👤 Student Workflow (/student1_1 to /student1_4)

- **Student Assistant (/student1_1):** AI chat box providing instant verified academic answers with confidence score and handbook citations.
- **Chat History (/student1_2):** Searchable history of previous questions asked by the student categorized by academic area.
- **Response Feedback (/student1_3):** Rating interface allowing students to submit thumbs up/down on AI response accuracy.
- **Student Profile (/student1_4):** Academic standing overview, CGPA, and live 75% attendance threshold tracker.

👤👤 Parent Workflow (/parent/dashboard to /parent/profile)

- **Parent Assistant (/parent/dashboard):** Context banner linked to specific ward (Rahul K, Sanjiv, Sujan) with tailored suggested chips.
- **Parent History (/parent/history):** Dedicated query log displaying fee deadlines, attendance policies, and hostel late permit rules.
- **Parent Feedback (/parent/feedback):** Report incorrect or missing AI information directly to trigger Faculty Review.
- **Parent Profile (/parent/profile):** Guardian details alongside linked ward academic summary and standing arrears check.

👤👤 Faculty Workflow (/faculty/dashboard to /faculty/draft)

- **Faculty Dashboard (/faculty/dashboard):** Overview of department queries (2,450 total), 820 students, 92% AI search accuracy, and assigned queue.
- **Assigned Queries (/faculty/queries):** Queue of low-confidence student queries explicitly assigned to Dr. Arun Kumar or Dr. Sathishkumar by Admin.
- **Submit FAQ Draft (/faculty/draft):** Form to compose expert faculty answers and send them for Admin approval.
- **Department Analytics (/faculty/analytics):** Query volume trends, top asked topics breakdown, and resolution speed analytics.

👤👤 Admin Workflow (/admin1 to /admin3_2)

- **Admin Dashboard (/admin1):** Control panel with system metrics, active users, vector store health, and continuous improvement pipeline.
- **User Management (/admin2):** User roster table with Create User modal, Edit Role modal, Active/Disable status toggle, and filters.
- **FAQ Approvals & Assign Faculty (/admin3):** Assign specific faculty (Dr. Arun Kumar [FAC001] or Dr. Sathishkumar [FAC002]) to FAQs and approve vector DB publishing.
- **Analytics Monitoring (/admin3_1):** Comprehensive query volume breakdown across all departments and unresolved query rates.
- **RAG & Vector DB Settings (/admin3_2):** Configure Sentence Transformer embeddings, FAISS vector store, similarity threshold (0.75), and top-K.